Mandatory Assignment 1

GFS189, RTL959

March 06, 2026

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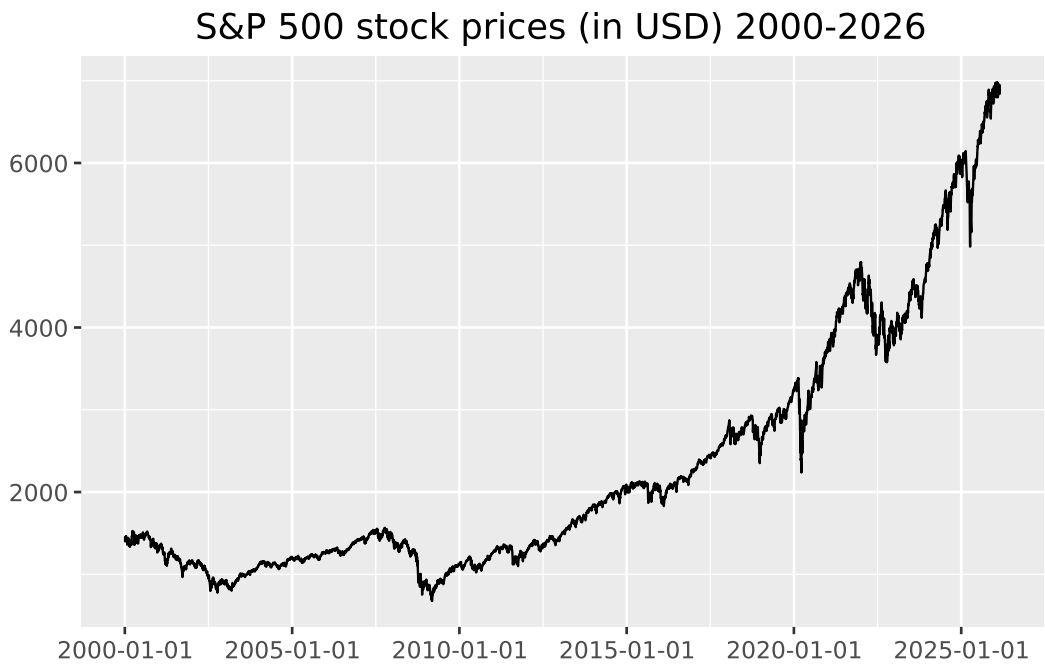
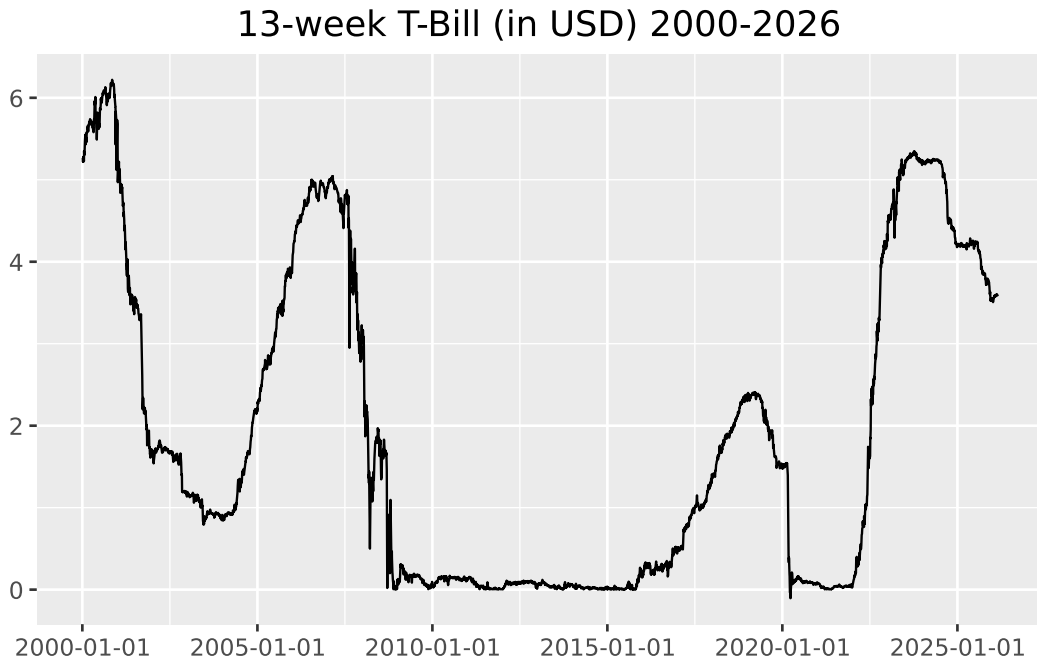
In this assignment we are investigating how different portfolio weights can lead to statistical errors based on the paper of Brandt(2010) and Jobson et.al(1980). The method to test this is to check how our portfolio weights deviate from the optimal in a single factor model market model which treats the calibrated parameters as the true values.

1





1. Visualize the distribution of the daily returns.

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Monthly mean excess return:       0.006
Monthly variance:                 0.002
Annualized mean excess return:    0.071
Annualized variance:              0.023
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2

3

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Tangency portfolio Sharpe: 0.1335
Tangency mu (monthly):     0.0063
Tangency sigma (monthly):  0.0473
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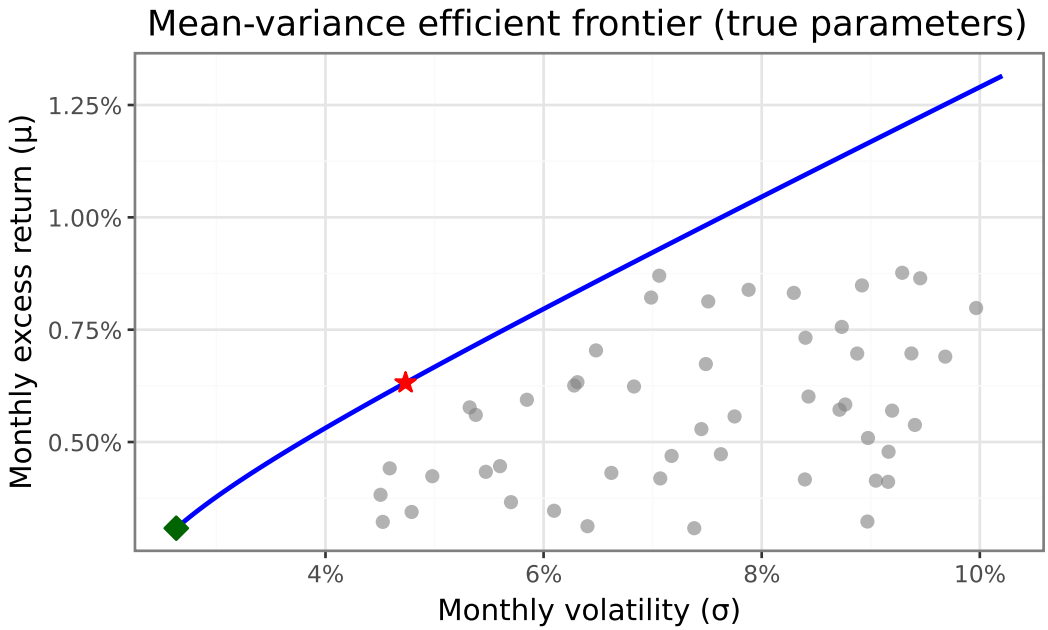
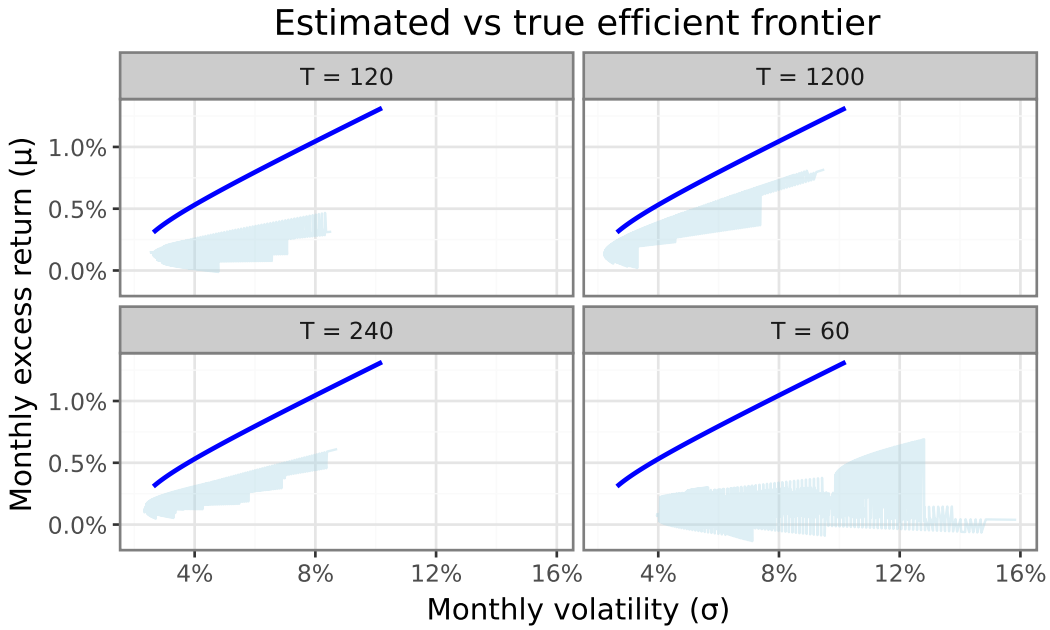
4



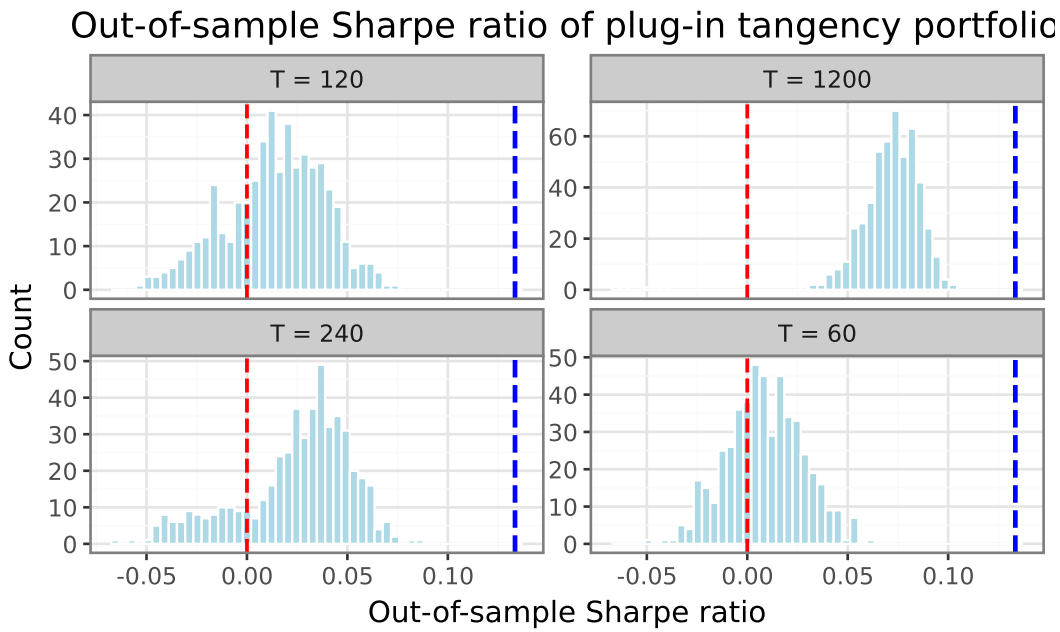
gency portfolio. Green diamond: minimum variance portfolio. Grey dots: individual assets.

5



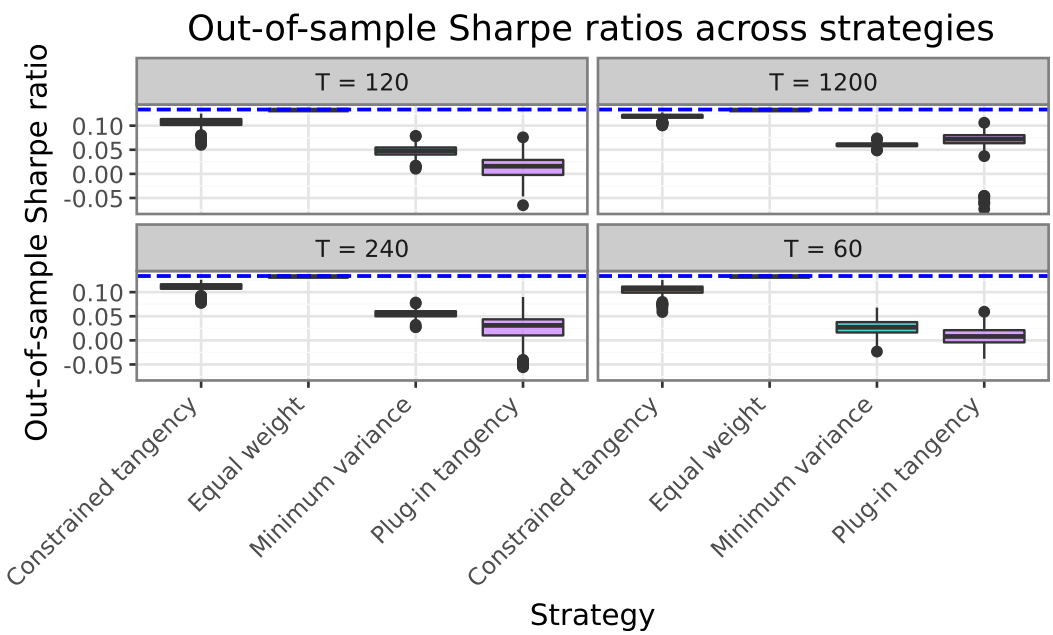
Light blue: plug-in frontiers. Dark blue: true frontier.

6



Dashed blue: true tangency Sharpe. Dashed red: zero.

7



Dashed blue: true tangency Sharpe.

References